

C-FLOX 250 & 500 TABLETS

Description: C-flox 250: White to off white coloured, round, biconvex, film coated tablet plain on one side and bisecting line on other side of the tablet.

Clinical Pharmacology: Ciprofloxacin exhibits potent bactericidal action by interfering with enzyme DNA gyrase - which is needed for the synthesis of bacterial DNA. The in vitro antimicrobial spectrum of Ciprofloxacin includes wide range of microorganisms including Gram-negative bacteria like campylobacter jejuni, E.coli, Citrobacters, H.influenzae, H.parainfluenzae, Shigella, Salmonella typhi, Neisseria-gonorrhoea, N.meningitidis, Branhamella catanhalis, Kleb.pneumoniae, Brucella melitensis, H.ducruyi, Proteus vulgaris, Pseudomonas aeruginosa, Providencia rettgeri, P. stuartii, Serratia marcescens, Enterobacter cloacae, and Gram-positive staphylococcus aureus, Staph. epidermidis, Staph.haemolyticus, Streptococcus pneumoniae, Streptococcus pyogenes. ... Most strains of P. multocipha and most anaerobic bacteria are resistant to Ciprofloxacin.Ciprofloxacin is slightly less active at acidic pH. It does not cross- react with other antimicrobial agents. Synergism is reported with aminoglycosides, clindamycin, metronidazole and t-lactams. Ciprofloxacin is also shown to have post antibiotic effect (PAE) i.e. continued antibacterial effect after cessation of drug. Ciprofloxacin is rapidly and well absorbed from the gastrointestinal tract. Oral bioavailability is about 70% and a peak plasma level of about 2.5 mcg per ml is achieved 1 to 2 hours after a dose of 500 mg by mouth. Absorption is delayed in presence of food. Plasma half-life ranges from 20 to 40 min. Ciprofloxacin is widely distributed in the body and tissue penetration is generally good. High concentrations are achieved in bile. The plasma half-life is about 3.5 to 4.5 hours, and there is modest accumulation. Half life may be prolonged in severe renal failure. Ciprofloxacin is eliminated principally by urinary excretion and non-renal clearance accounts for one third of its elimination. Urinary excretion is tubular secretion and by glomerular filtration. Urinary excretion is reduced by Probenecid. Atleast 4 active metabolites are identified. Oxociprofloxacin is major urinary metabolite and sulpiciprofloxacin the primary fecal metabolite. About 40 to 50% of an oral dose is excreted in the urine and about 15% as metabolites.

According to in-vitro investigations, the following pathogens can be regarded as sensitive: *E.Coli*, *Shigella*, *Salmonella*, *Citrobacter*, *Klebsiella*, *Enterobacter*, *Serratia*, *Hafnia*, *Edwardsiella*, *Proteus* (indole-positive and indole-negative), *Providencia*, *Morganella*, *Yersinia*; *Vibrio*, *Aeromonas*, *Plesiomonas*, *Pasteurella*, *Haemophilus*, *Campylobacter*, *Pseudomonas*, *Legionella*, *Nisseria*, *Moraxella*, *Acinetobacter*, *Brucella*; *Staphylococcus*, *Listeria*, *Corynebacterium*, *Chlamydia*.

The following are usually resistant: *Enterococcus faecium*, *Ureaplasma urealyticum*, *Nocardia asteroides*. With a few exceptions anaerobes are moderately sensitive, e.g. *Peptococcus*, *Peptostreptococcus* to resistant e.g. *Bacteroides*. Ciprofloxacin is ineffective against *Treponema pallidum*.

Route and Method of Administration: Oral. The tablets are swallowed whole with a small amount of fluid. They can be taken independent of mealtimes, (if the tablets are taken on an empty stomach, the active substance is absorbed more rapidly). If the patient is unable to take tablets, because of the severity of the illness or for other reasons, it is recommended to commence the therapy with an intravenous form of ciprofloxacin. After intravenous administration the treatment can be continued orally.

Elderly: Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance. **Children:** Contraindicated. **Renal and Hepatic Impairment:** 1. Impaired renal function: 1.1 Where creatinine clearance is between 31 and 60 ml/min/1.73m² or where the serum creatinine concentration is between 1.4 and 1.9 mg/100 ml, the maximum daily dose should be 1000 mg per day for oral administration

Contraindications: Ciprofloxacin is contraindicated in cases of hypersensitivity to Ciprofloxacin or other quinolone chemotherapeutics. Ciprofloxacin must not be prescribed for children, adolescents, since there is no experience on the drug's safety in these patients groups and since, on the basis of animal studies, it is not entirely improbable that the drug could cause damage to articular cartilage in the immature organism.

Nervous System: In patients of epilepsy and those having history of CNS-disorders (eg: lowered convulsion threshold, previous history of convulsion, reduced cerebral blood flow, altered brain structure or stroke), ciprofloxacin should only be used where the benefits of treatment exceed the risks, since these patients are endangered because of possible central-nervous side effects. In some instances the CNS reactions occurred after the first administration of ciprofloxacin. In rare cases depression or psychosis can progress to self endangering behaviour. In these cases ciprofloxacin should be discontinued and the doctor should be informed immediately.



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